Week 3

Linear transformations and linear functionals:
Definition 2.21: A function $T : \mathbb{R}^n \to \mathbb{R}^m$ / $T : \mathbb{R}^n \to \mathbb{R}$ is called a *linear transformation* / *linear functional* if the following **linearity** *axiom* holds for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ and all $\lambda_1, \lambda_2 \in \mathbb{R}$:

$$T(\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2) = \lambda_1 T(\mathbf{x}_1) + \lambda_2 T(\mathbf{x}_2).$$

The same can be said with two axioms (Lemma 2.23): For all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$ and all $\lambda \in \mathbb{R}$,

(i) $T(\mathbf{x} + \mathbf{x}') = T(\mathbf{x}) + T(\mathbf{x}')$, and

(ii) $T(\lambda\mathbf{x}) = \lambda T(\mathbf{x})$.

Observation 2.22: Every matrix transformation is a linear transformation.

Proof. Use $T_A(\mathbf{x}) = A\mathbf{x}$ and Lemma 2.19:

$$\begin{tikzcd} \mathbf{x}_1, \mathbf{x}_2 \arrow[r, "\text{linear combination}"] \arrow[d, "T_A"'] & \lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 \arrow[d, "T_A"] \\ T_A(\mathbf{x}_1), T_A(\mathbf{x}_2) \arrow[r, "\text{linear combination}"'] & \lambda_1 T_A(\mathbf{x}_1) + \lambda_2 T_A(\mathbf{x}_2) = T_A(\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2) \end{tikzcd}$$

□

Examples:

1. Let $\mathbf{v} \in \mathbb{R}^m$. $T : \mathbb{R}^m \to \mathbb{R}$,

$$T : \mathbf{x} \mapsto \sum_{i=1}^{m} v_i x_i = \mathbf{v}^\top \mathbf{x}$$

is a linear functional (the covector $\mathbf{v}^\top$; see Definition 1.20). For the proof, verify (i) and (ii) in Lemma 2.23:

$$\begin{aligned} T(\mathbf{x} + \mathbf{x}') &= \sum_{i=1}^{m} v_i(x_i + x_i') = \sum_{i=1}^{m} v_i x_i + \sum_{i=1}^{m} v_i x_i' = T(\mathbf{x}) + T(\mathbf{x}'), \text{ and} \\ T(\lambda\mathbf{x}) &= \sum_{i=1}^{m} v_i(\lambda x_i) = \lambda \sum_{i=1}^{m} v_i x_i = \lambda T(\mathbf{x}). \end{aligned}$$

2. $T : \mathbb{R}^3 \to \mathbb{R}^2$,

$$T : \mathbf{x} \to \begin{pmatrix} x_1 + 2x_2 \\ 3x_2 + 4x_3 \end{pmatrix}$$

is a linear transformation. For the proof, observe that $T = T_A$ for

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \end{bmatrix} : \quad T_A(\mathbf{x}) = A\mathbf{x} = \begin{pmatrix} 1x_1 + 2x_2 + 0x_3 \\ 0x_1 + 3x_2 + 4x_3 \end{pmatrix} = T(\mathbf{x}).$$

Now use Observation 2.22 (every matrix transformation is a linear transformation).

3. $T : \mathbb{R}^n \to \mathbb{R}^m, \mathbf{x} \mapsto \mathbf{0}$ is a linear transformation ($T = T_A$ for $A = 0$).

4. $T : \mathbb{R}^m \to \mathbb{R}^m, \mathbf{x} \mapsto \mathbf{x}$ is a linear transformation ($T = T_A$ for $A = I$).

5. $T : \mathbb{R}^m \to \mathbb{R}$,

$$T : \mathbf{x} \mapsto \sum_{i=1}^{m} |x_i| = \|\mathbf{x}\|_1 \quad \text{(1-norm of } \mathbf{x}\text{)}$$

is not a linear functional: if $\mathbf{x} \neq \mathbf{0}$ and $\lambda < 0$, then $T(\lambda\mathbf{x}) > 0$ but $\lambda T(\mathbf{x}) < 0$; so (ii): $T(\lambda\mathbf{x}) = \lambda T(\mathbf{x})$, fails.

6. $T : \mathbb{R}^n \to \mathbb{R}^m, \mathbf{x} \mapsto \mathbf{v} \neq 0$ is not a linear transformation: (i): $T(\mathbf{x}+\mathbf{x}') = T(\mathbf{x})+T(\mathbf{x}')$, fails with $T(\mathbf{x}+\mathbf{x}') = \mathbf{v}$ and $T(\mathbf{x}) + T(\mathbf{x}') = 2\mathbf{v} \neq \mathbf{v}$.

"Null in, null out":
Lemma 2.24: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ / $T : \mathbb{R}^n \to \mathbb{R}$ be a linear transformation / linear functional. Then $T(\mathbf{0}) = \mathbf{0}$ / 0.

Proof. Use (ii), $T(\lambda\mathbf{x}) = \lambda T(\mathbf{x})$: $T(\mathbf{0}) = T(0\mathbf{x}) = 0T(\mathbf{x}) = \mathbf{0}$ / 0. □

"Linearity with more terms":
Lemma 2.25: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ / $T : \mathbb{R}^n \to \mathbb{R}$ be a linear transformation / linear functional, let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_\ell \in \mathbb{R}^n$ and $\lambda_1, \lambda_2, \ldots, \lambda_\ell \in \mathbb{R}$. Then

$$T\left(\sum_{j=1}^{\ell} \lambda_j \mathbf{x}_j\right) = \sum_{j=1}^{\ell} \lambda_j T(\mathbf{x}_j).$$

$\ell = 0$: Lemma 2.24 $\qquad$ $\ell = 1$: (ii) $\qquad$ $\ell = 2$: Definition 2.21 $\qquad$ $\ell > 2$?

Proof.

$$T\left(\sum_{j=1}^{\ell} \lambda_j \mathbf{x}_j\right) = T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j + \lambda_\ell \mathbf{x}_\ell\right) \overset{(i)}{=} T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j\right) + T\left(\lambda_\ell \mathbf{x}_\ell\right) \overset{(ii)}{=} T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j\right) + \lambda_\ell T\left(\mathbf{x}_\ell\right).$$

Same for $\ell - 1$:

$$= T\left(\sum_{j=1}^{\ell-2} \lambda_j \mathbf{x}_j\right) + \lambda_{\ell-1} T\left(\mathbf{x}_{\ell-1}\right) + \lambda_\ell T\left(\mathbf{x}_\ell\right).$$

Repeating for $\ell - 2, \ldots, 1$:

$$T\left(\sum_{j=1}^{\ell} \lambda_j \mathbf{x}_j\right) = \underbrace{T\left(\sum_{j=1}^{0} \lambda_j \mathbf{x}_j\right)}_{T(\mathbf{0})=\mathbf{0}\ /\ 0} + \lambda_1 T(\mathbf{x}_1) + \ldots + \lambda_{\ell-1} T\left(\mathbf{x}_{\ell-1}\right) + \lambda_\ell T\left(\mathbf{x}_\ell\right) = \sum_{j=1}^{\ell} \lambda_j T(\mathbf{x}_j).$$

$\square$

Proof by induction (without "repeating"):
For $\ell = 0$, prove it directly (*base case*): statement reads as $T(\mathbf{0}) = \mathbf{0}\ /\ 0$ which is Lemma 2.24.
For $\ell > 0$, prove the *induction step*: if the statement is true for $\ell - 1$ (*induction hypothesis*), then it is also true for ℓ.
Having done this, we know that the statement is true for all ℓ. Induction step:

$$\begin{aligned}
T\left(\sum_{j=1}^{\ell} \lambda_j \mathbf{x}_j\right) &= T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j + \lambda_\ell \mathbf{x}_\ell\right) \stackrel{(i)}{=} T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j\right) + T\left(\lambda_\ell \mathbf{x}_\ell\right) \\
&\stackrel{(ii)}{=} T\left(\sum_{j=1}^{\ell-1} \lambda_j \mathbf{x}_j\right) + \lambda_\ell T\left(\mathbf{x}_\ell\right) \quad \text{(as before)} \\
&= \sum_{j=1}^{\ell-1} \lambda_j T(\mathbf{x}_j) \quad + \lambda_\ell T\left(\mathbf{x}_\ell\right) \quad \text{(induction hypothesis)} \\
&= \sum_{j=1}^{\ell} \lambda_j T(\mathbf{x}_j).
\end{aligned}$$

$\square$

The matrix of a linear transformation:
Theorem 2.26: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. There exists a unique $m \times n$ matrix A such that $T = T_A$.

Proof. For $T = T_A$, we need $T(\mathbf{e}_j) = T_A(\mathbf{e}_j) = A\mathbf{e}_j$ (j-th column of A), for all $j \in [n]$. Only candidate is

$$A = \begin{bmatrix} | & | & & | \\ T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \\ | & | & & | \end{bmatrix}.$$

This works:

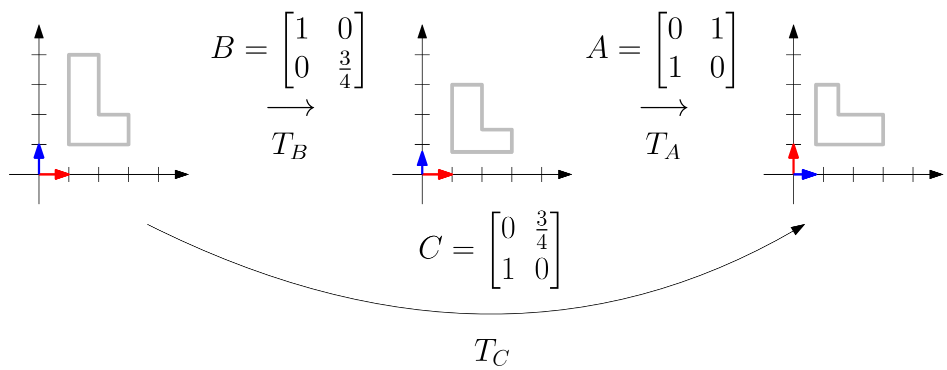
$$\begin{aligned}
T_A(\mathbf{x}) &= A\mathbf{x} && \text{(Definition 2.18, } T_A\text{)}\\
&= \sum_{j=1}^{n} x_j T(\mathbf{e}_j) && \text{(Definition 2.4, matrix-vector multiplication)}\\
&= T\left(\sum_{j=1}^{n} x_j \mathbf{e}_j\right) && \text{(Lemma 2.25)}\\
&= T(\mathbf{x}) && \text{(as in } \begin{pmatrix} x_1\\ x_2\\ x_3\end{pmatrix} = x_1 \underbrace{\begin{pmatrix}1\\0\\0\end{pmatrix}}_{\mathbf{e}_1} + x_2 \underbrace{\begin{pmatrix}0\\1\\0\end{pmatrix}}_{\mathbf{e}_2} + x_3 \underbrace{\begin{pmatrix}0\\0\\1\end{pmatrix}}_{\mathbf{e}_3}\text{)}
\end{aligned}$$

□

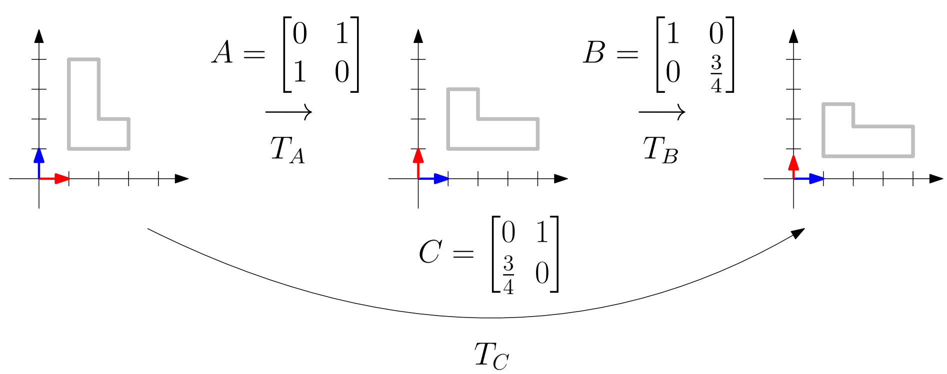
Consequence: If we know $T(\mathbf{e}_1), T(\mathbf{e}_2), \ldots, T(\mathbf{e}_n)$, we know A and $T(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$.

Matrix multiplication (Section 2.3)

Combining two matrix transformations…



"Stretching, then mirroring"



"Mirroring, then stretching"

…gives another matrix transformation (columns of C: what happens to $\mathbf{e}_1, \mathbf{e}_2$ in the end?)

Definition 2.33 (Composition of functions): Let $g : X \to Y$ and $f : Y \to Z$ be two functions where X, Y, Z are arbitrary sets. The function $h : X \to Z$,

$$h : x \mapsto f(g(x))$$

is the *composition* of f and g, written as $h = f \circ g$ ("first apply g, then f").

Lemma 2.34 (Composition of matrix transformations): Let $T_B : \mathbb{R}^b \to \mathbb{R}^n$ and $T_A : \mathbb{R}^n \to \mathbb{R}^a$ be two matrix transformations. The composition $T_A \circ T_B : \mathbb{R}^b \to \mathbb{R}^a$ ("first do T_B, then T_A") is another matrix transformation.

Proof. Step 1: $T_A \circ T_B$ is a linear transformation (Definition 2.21):

$$\begin{aligned}
(T_A \circ T_B)(\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2) &= T_A(T_B(\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2)) \text{ (composition)} \\
&= T_A(\lambda_1 T_B(\mathbf{x}_1) + \lambda_2 T_B(\mathbf{x}_2)) \text{ (linearity of } T_B) \\
&= \lambda_1 T_A(T_B(\mathbf{x}_1)) + \lambda_2 T_A(T_B(\mathbf{x}_2)) \text{ (linearity of } T_A) \\
&= \lambda_1 (T_A \circ T_B)(\mathbf{x}_1) + \lambda_2 (T_A \circ T_B)(\mathbf{x}_2). \text{ (composition)}
\end{aligned}$$

Step 2: $T_A \circ T_B$ is a matrix transformation by Lemma 2.25. □

What is the matrix of $T_A \circ T_B$?
Lemma 2.35: Let A be an $a \times n$ matrix ($T_A : \mathbb{R}^n \to \mathbb{R}$)
and

$$B = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_b \\ | & | & & | \end{bmatrix}$$

an $n \times b$ matrix. ($T_B : \mathbb{R}^b \to \mathbb{R}^n$)
The $a \times b$ matrix ($T_A \circ T_B : \mathbb{R}^b \to \mathbb{R}^a$)

$$C = \begin{bmatrix} | & | & & | \\ A\mathbf{x}_1 & A\mathbf{x}_2 & \cdots & A\mathbf{x}_b \\ | & | & & | \end{bmatrix}$$

is the unique matrix that satisfies $T_C = T_A \circ T_B$.

Proof. Both T_C and $T_A \circ T_B$ must produce the same output on every standard unit vector:

$$T_C(\mathbf{e}_j) = (T_A \circ T_B)(\mathbf{e}_j) \stackrel{\text{Def. 2.33}}{=} T_A(T_B(\mathbf{e}_j)) \stackrel{\text{Def. 2.18}}{=} AT_B(\mathbf{e}_j), \quad j = 1, 2, \ldots, b.$$

$T_C(\mathbf{e}_j) = C\mathbf{e}_j$ (j-th column of C) $T_B(\mathbf{e}_j) = B\mathbf{e}_j$ (j-th column of B: $\mathbf{x}_j$) .
So, previous derivation says: j-th column of C is $A\mathbf{x}_j$. □

Let's call C the *product* of A and B:

Definition 2.36 (Matrix multiplication in column notation)
Let A be an $a \times n$ matrix and

$$B = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_b \\ | & | & & | \end{bmatrix}$$

an $n \times b$ matrix. The $a \times b$ matrix

$$AB := \begin{bmatrix} | & | & & | \\ A\mathbf{x}_1 & A\mathbf{x}_2 & \cdots & A\mathbf{x}_b \\ | & | & & | \end{bmatrix}$$

is the *product* of A and B.

Corollary 2.37: Let A be an $a \times n$ matrix and B be an $n \times b$ matrix. Then $T_A \circ T_B = T_{AB}$.

$$\begin{tikzcd} A, B \arrow[r, "\text{product}"] \arrow[d, "\text{construct transformation}"'] & AB \arrow[d, "\text{construct transformation}"] \\ T_A, T_B \arrow[r, "\text{composition}"'] & T_A \circ T_B \end{tikzcd}$$

"Usual" definition of AB: rows of A times columns of B

Observation 2.38: Let

$$A = \begin{bmatrix} — & \mathbf{u}_1^\top & — \\ — & \mathbf{u}_2^\top & — \\ & \vdots & \\ — & \mathbf{u}_a^\top & — \end{bmatrix} \in \mathbb{R}^{a \times n}, \quad B = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_b \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times b}.$$

Then

$$AB = \underbrace{\begin{bmatrix} \mathbf{u}_1^\top \mathbf{x}_1 & \mathbf{u}_1^\top \mathbf{x}_2 & \cdots & \mathbf{u}_1^\top \mathbf{x}_b \\ \mathbf{u}_2^\top \mathbf{x}_1 & \mathbf{u}_2^\top \mathbf{x}_2 & \cdots & \mathbf{u}_2^\top \mathbf{x}_b \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_a^\top \mathbf{x}_1 & \mathbf{u}_a^\top \mathbf{x}_2 & \cdots & \mathbf{u}_a^\top \mathbf{x}_b \end{bmatrix}}_{ab \text{ scalar products}} = \left[\mathbf{u}_i^\top \mathbf{x}_j\right]_{i=1,j=1}^{a \quad b} \in \mathbb{R}^{a \times b}.$$

Works because $A\mathbf{x}_j$ can be written as $(\mathbf{u}_i^\top \mathbf{x}_j)_{i=1}^a$ (Observation 2.8). Example:

$$\begin{bmatrix} \mathbf{2} & \mathbf{3} \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -3 & \mathbf{1} & 3 \\ 2 & -\mathbf{1} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\mathbf{1} & 6 \\ -11 & 4 & 9 \end{bmatrix} \qquad \begin{bmatrix} 2 & 3 \\ \mathbf{3} & -\mathbf{1} \end{bmatrix} \begin{bmatrix} -3 & 1 & \mathbf{3} \\ 2 & -1 & \mathbf{0} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 6 \\ -11 & 4 & \mathbf{9} \end{bmatrix}$$

$$\mathbf{2} \cdot \mathbf{1} + \mathbf{3} \cdot (-\mathbf{1}) = -\mathbf{1} \qquad \mathbf{3} \cdot \mathbf{3} + (-\mathbf{1}) \cdot \mathbf{0} = \mathbf{9}$$

Previous examples:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ (mirroring)}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & \frac{3}{4} \end{bmatrix} \text{ (stretching)}$$

$$AB = \begin{bmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \frac{3}{4} \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ \frac{3}{4} \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 0 & \frac{3}{4} \\ 1 & 0 \end{bmatrix}, \quad BA = \begin{bmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 0 & \frac{3}{4} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \begin{pmatrix} 0 & \frac{3}{4} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{3}{4} & 0 \end{bmatrix}.$$

"Stretching, then mirroring" "Mirroring, then stretching"

Corollary 2.41: Let I be the $m \times m$ identity matrix. Then $IA = A$ for all $m \times n$ matrices, and $AI = A$ for all $n \times m$ matrices.

Lemma 2.40: Let A be an $a \times n$ matrix and B an $n \times b$ matrix. Then

$$(AB)^\top = B^\top A^\top.$$

Works for more matrices, for example $(ABC)^\top = C^\top B^\top A^\top$.

Distributivity and associativity:
Lemma 2.42:

(i) $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$; (*distributivity*, easy to check)

(ii) $(AB)C = A(BC)$. (*associativity*)

Proof of associativity. By Corollary 2.37:

- $(AB)C$ is the matrix of $(T_A \circ T_B) \circ T_C$.
- $A(BC)$ is the matrix of $T_A \circ (T_B \circ T_C)$.
- $(T_A \circ T_B) \circ T_C$ and $T_A \circ (T_B \circ T_C)$ are the same linear transformations (and therefore have the same matrices $(AB)C = A(BC)$ by Theorem 2.26):

$$\begin{aligned} ((T_A \circ T_B) \circ T_C)(\mathbf{x}) &= (T_A \circ T_B)(T_C(\mathbf{x})) = T_A(T_B(T_C(\mathbf{x}))), \\ (T_A \circ (T_B \circ T_C))(\mathbf{x}) &= T_A((T_B \circ T_C)(\mathbf{x})) = T_A(T_B(T_C(\mathbf{x}))). \end{aligned}$$

□

We can omit the brackets: ABC.

Also works for more matrices with the same proof (*generalized associativity*). For example,

$$(AB)(CD) = A((BC)D) = \cdots = ABCD.$$